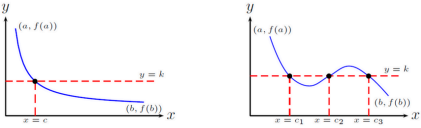


1.8 Intermediate Value Theorem (IVT)

If a real-valued function f is continuous on $[a, b]$ and k is a number between $f(a)$ and $f(b)$, then $f(c) = k$ for some $c \in [a, b]$.



Example 1.15. Show that the equation $x^3e^x = 10$ has a solution between 1 and 1.5.

Solution. Let $f(x) = x^3e^x$. f is continuous on $\mathbb{R}$. We have $f(1) = e = 2.718$ and $f(1.5) = 1.5^3e^{1.5} = 15.126$. Thus $f(1) < 10 < f(1.5)$. By Intermediate Value Theorem, there is a solution to $f(x) = 10$ between 1 and 1.5.

Result. Limits of the form $\lim_{x \rightarrow \pm\infty} \frac{P(x)}{Q(x)}$, where $P(x)$ and $Q(x)$ are polynomials in x .

$$\lim_{x \rightarrow \pm\infty} \frac{P(x)}{Q(x)} = \lim_{x \rightarrow \pm\infty} \underbrace{\frac{Ax^\alpha}{Bx^\beta}}_{\text{leading term}} + \dots = \begin{cases} 0 & \text{if } \alpha < \beta \\ \frac{A}{B} & \text{if } \alpha = \beta \\ \infty \text{ or } -\infty & \text{if } \alpha > \beta \end{cases}$$

depends on the question

Useful results.

If $\lim_{x \rightarrow c} g(x) = 0$, then

- $\lim_{x \rightarrow c} \frac{\sin(g(x))}{g(x)} = \lim_{x \rightarrow c} \frac{g(x)}{\sin(g(x))} = 1,$
- $\lim_{x \rightarrow c} \frac{\tan(g(x))}{g(x)} = \lim_{x \rightarrow c} \frac{g(x)}{\tan(g(x))} = 1.$

Squeeze Theorem. Suppose $g(x) \leq f(x) \leq h(x)$ for all x in some open interval containing a point c , except possibly at $x = c$. If $\lim_{x \rightarrow c} g(x) = 0 = \lim_{x \rightarrow c} h(x)$ or $\lim_{x \rightarrow c} f(x) = L$, then $\lim_{x \rightarrow c} f(x) = L$.

Theorem 2.1. If f is differentiable at $x = x_0$, then f is continuous at $x = x_0$.

Theorem 3.10. Let f and g be differentiable at all points in some open interval containing $x = c$ (except possibly at c). If $\lim_{x \rightarrow c} f(x) = 0 = \lim_{x \rightarrow c} g(x)$ or $\lim_{x \rightarrow c} f(x) = \infty = \lim_{x \rightarrow c} g(x)$, then

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)},$$

provided the limit on the right exists or equals ∞ or $-\infty$.

The result also holds

- (1) for limits at infinity, i.e. $c = \pm\infty$.
- (2) for one-sided limits.

Theorem 3.11. (Rolle's Theorem) Let f be continuous on $[a, b]$ and differentiable on (a, b) . If $f(a) = f(b)$, then there is at least one number c in (a, b) such that $f'(c) = 0$.

Exercise 3.10. Use Rolle's Theorem to prove that the equation

$$2e^x + x^2 + 3x = 0$$

has at most two real solutions.

For $f(x) = 2e^x + x^2 + 3x$, Apply Rolle again to f' on $[x_1, x_2]$:

- $\exists x_3 \in (x_1, x_2)$ with $f''(x_3) = 0$.

But $a < b < c$, $f'(x) = 2e^x + 2x + 3$, $f''(x) = 2e^x + 2 > 0 \forall x$, so $f''(x_3) = 0$ is impossible.

Therefore, $f(x)$ cannot have 3 distinct real roots.

Thus f' has two distinct zeros $x_1 < x_2$. Any function with 4 roots must also have 3 roots, so this excludes 3, 4, 5, ... roots.

Theorem 3.12. (Mean Value Theorem) Let f be continuous on $[a, b]$ and differentiable on (a, b) . Then, there is at least one number c in (a, b) such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Exercise 3.11. Use the Mean Value Theorem to prove that for all real numbers x and y ,

$$|\cos x - \cos y| \leq |x - y|.$$

Claim: For all real x, y ,

$$|\cos x - \cos y| \leq |x - y|.$$

Let $f(t) = \cos t$. Then f is continuous and differentiable with $f'(t) = -\sin t$, so $|f'(t)| \leq 1$ for all t .

If $x = y$ the inequality is trivial. For $x \neq y$, by MVT there exists c between x and y such that

$$\frac{\cos x - \cos y}{x - y} = f'(c) = -\sin c.$$

Thus

$$|\cos x - \cos y| = |x - y| \cdot |\sin c| \leq |x - y| \cdot 1 = |x - y|.$$

Exercise 3.12. Use the Mean Value Theorem to show that if $f'(x) = 0$ for all x in (a, b) , then f is a constant function on (a, b) .

Claim: If $f'(x) = 0$ for all $x \in (a, b)$, then f is constant on (a, b) .

Take any $x_1 < x_2$ in (a, b) . By MVT, there exists $c \in (x_1, x_2)$ such that

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} = f'(c) = 0.$$

Hence $f(x_2) - f(x_1) = 0$, so $f(x_2) = f(x_1)$.

Since this holds for all $x_1, x_2 \in (a, b)$, f is constant on (a, b) .

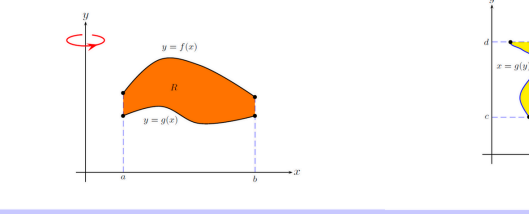
Theorem 3.4. (Extreme Value Theorem) If f is continuous on a closed interval $[a, b]$, then f has an absolute maximum and an absolute minimum at some points in $[a, b]$.

Types of functions	Examples	Remark
Logarithmic Function	$\ln(ax + b)$ or its higher powers	differentiate it
Inverse Trigonometric Functions	$\sin^{-1}(ax + b)$, $\cos^{-1}(ax + b)$, $\tan^{-1}(ax + b)$	differentiate it
Algebraic Functions	Power functions x^a , polynomials	differentiate it
Trigonometric Functions	$\sin(ax + b)$, $\cos(ax + b)$, $\tan(ax + b)$, $\csc(ax + b)$, $\sec(ax + b)$, $\cot(ax + b)$, or a combinations of these	differentiate it integrate it
Exponential Functions	e^{ax+b}	integrate it

Riemann Sum: Approximation. For sufficiently large n ,

$$\int_a^b f(x) dx \approx \sum_{k=1}^n \left(\frac{b-a}{n} \right) f\left(a + k\left(\frac{b-a}{n}\right)\right).$$

Theorem 5.10. Let f and g be continuous on $[c, d]$ with $f(y) \geq g(y) \geq 0$ for all $c \leq y \leq d$. When the region bounded by the curves $x = f(y)$ and $x = g(y)$ for $c \leq y \leq d$ is revolved completely about the y -axis, the volume of the solid formed is

$$V = \pi \int_c^d f(y)^2 dy - \pi \int_c^d g(y)^2 dy.$$


Theorem 5.12. When the plane region bounded by the curve $y = f(x)$, $y = g(x)$ and the lines $x = a$ and $x = b$, where $0 \leq a < b$, is revolved completely about the y -axis, the volume of the solid formed is

$$V = 2\pi \int_a^b x(f(x) - g(x))dx.$$

Theorem 5.14. When the plane region bounded by the curve $x = f(y)$, $x = g(y)$ and the lines $y = c$ and $y = d$, where $0 \leq c < d$, is revolved completely about the x -axis, the volume of the solid formed is

$$V = 2\pi \int_c^d y(f(y) - g(y))dy.$$

Theorem 6.2. (Squeeze Theorem for Sequence) If $a_n \leq b_n \leq c_n$ for all n and $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L$, then $\lim_{n \rightarrow \infty} b_n = L$.

If $\{a_n\}$ and $\{b_n\}$ are convergent sequences and c is a constant, then we have

- $\lim_{n \rightarrow \infty} ca_n = c \lim_{n \rightarrow \infty} a_n$.
- $\lim_{n \rightarrow \infty} (a_n \pm b_n) = \lim_{n \rightarrow \infty} a_n \pm \lim_{n \rightarrow \infty} b_n$.
- $\lim_{n \rightarrow \infty} a_n b_n = \lim_{n \rightarrow \infty} a_n \cdot \lim_{n \rightarrow \infty} b_n$.
- $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{\lim_{n \rightarrow \infty} a_n}{\lim_{n \rightarrow \infty} b_n}$, if $\lim_{n \rightarrow \infty} b_n \neq 0$.

Theorem 6.5. (The n^{th} Term Test for Divergence)

If $\lim_{n \rightarrow \infty} a_n$ does not exist or if $\lim_{n \rightarrow \infty} a_n \neq 0$, then the series $\sum_{n=1}^{\infty} a_n$ is divergent.

Theorem 6.6. A series $\sum_{n=1}^{\infty} a_n$ of nonnegative terms converges if and only if its partial sums are bounded from above (i.e., there exists a constant K such that $S_n < K$ for all n .)

Theorem 6.7. (Integral Test) Let $\{a_n\}$ be a sequence of positive terms. Suppose that $a_n = f(n)$, where f is a continuous, positive, decreasing function of x for all $x \geq 1$. Then the series $\sum_{n=1}^{\infty} a_n$ is convergent if and only if the improper integral $\int_1^{\infty} f(x) dx$ is convergent. In other words:

- (i) If $\int_1^{\infty} f(x) dx$ is convergent, then $\sum_{n=1}^{\infty} a_n$ is convergent.
- (ii) If $\int_1^{\infty} f(x) dx$ is divergent, then $\sum_{n=1}^{\infty} a_n$ is divergent.

Theorem 6.8. (The p -series) The p -series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is convergent if and only if $p > 1$.

Theorem 6.9. (Comparison Test)

Suppose $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are series with nonnegative terms such that $0 \leq a_n \leq b_n$ for all n .

- (i) If $\sum_{n=1}^{\infty} b_n$ is convergent, then $\sum_{n=1}^{\infty} a_n$ is convergent.
- (ii) If $\sum_{n=1}^{\infty} a_n$ is divergent, then $\sum_{n=1}^{\infty} b_n$ is divergent.

Theorem 6.10. (The Ratio Test) Suppose $\sum_{n=1}^{\infty} a_n$ is a series such that $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L$ (L is a finite number or ∞).

- (i) If $0 \leq L < 1$, then $\sum_{n=1}^{\infty} a_n$ is absolutely convergent. That is $\sum_{n=1}^{\infty} |a_n|$ is convergent.
- (ii) If $L > 1$, then $\sum_{n=1}^{\infty} a_n$ is divergent.
- (iii) If $L = 1$, then the ratio test is inconclusive.

Theorem 6.11. (The Root Test) Suppose $\sum_{n=1}^{\infty} a_n$ is a series such that $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = L$ (L is a finite number or ∞).

- (i) If $0 \leq L < 1$, then $\sum_{n=1}^{\infty} a_n$ is absolutely convergent.
- (ii) If $L > 1$, then $\sum_{n=1}^{\infty} a_n$ is divergent.
- (iii) If $L = 1$, then the root test is inconclusive.

Theorem 6.12. (The Alternating Series Test) If b_n is a sequence of positive numbers such that

- (i) b_n is decreasing (that is, $b_n \geq b_{n+1}$ for all n), and
- (ii) $\lim_{n \rightarrow \infty} b_n = 0$,

then the alternating series

$$\sum_{n=1}^{\infty} (-1)^{n-1} b_n = b_1 - b_2 + b_3 - b_4 + \dots \quad \text{and} \quad \sum_{n=1}^{\infty} (-1)^n b_n = -b_1 + b_2 - b_3 + b_4 - \dots$$

are convergent.

Theorem 6.13. If $\sum_{n=1}^{\infty} |a_n|$ is convergent, then $\sum_{n=1}^{\infty} a_n$ is convergent.

Theorem 6.14. For a given power series $\sum_{n=0}^{\infty} c_n(x-a)^n$, exactly one of the following possibilities holds:

- (i) The series converges at $x = a$ only.
 - (ii) The series converges for all x .
 - (iii) There is a positive number R such that the series converges **absolutely** if $|x-a| < R$ and diverges if $|x-a| > R$.
- Theorem 6.15.** Consider the power series

Theorem 6.16. If the power series $\sum_{n=0}^{\infty} c_n(x-a)^n$ has radius of convergence $R > 0$, then the function f defined by

$$f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n$$

is differentiable on the interval $|x-a| < R$ and

$$(i) \quad f'(x) = \sum_{n=1}^{\infty} n c_n (x-a)^{n-1}, \text{ for } |x-a| < R.$$

$$(ii) \quad \int f(x) dx = \sum_{n=0}^{\infty} c_n \frac{(x-a)^{n+1}}{n+1} + C, \text{ for } |x-a| < R.$$

Theorem 6.17. If f has a power series representation at $x = a$, that is

$$f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n, \quad |x-a| < R, \text{ for some } R > 0,$$

then its coefficients are given by the formula

$$c_n = \frac{f^{(n)}(a)}{n!}.$$

If f has a power series representation at $x = a$, then it is unique and has the form

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n.$$

This is called the Taylor series of f at $x = a$.

The Maclaurin series of f is the special case of Taylor series when $a = 0$:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n.$$

Theorem 7.6 (Properties of Dot Product).

For vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ and any scalar d ,

- (i) $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$ (commutativity)
- (ii) $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$ (distributive law)
- (iii) $(d\mathbf{a}) \cdot \mathbf{b} = d(\mathbf{a} \cdot \mathbf{b}) = \mathbf{a} \cdot (d\mathbf{b})$
- (iv) $\mathbf{0} \cdot \mathbf{a} = 0$
- (v) $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2$.

Theorem 7.9 (Distance from a point to a plane). The (shortest) distance from a point $P(x_0, y_0, z_0)$ to the plane $ax + by + cz = d$ is given by

$$\frac{|ax_0 + by_0 + cz_0 - d|}{\sqrt{a^2 + b^2 + c^2}}.$$

Theorem 7.11. If θ is the angle between $\mathbf{a}$ and $\mathbf{b}$, then

$$|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}|\sin\theta.$$

$$(i) \quad \mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$$

$$(ii) \quad (d\mathbf{a}) \times \mathbf{b} = d(\mathbf{a} \times \mathbf{b}) = \mathbf{a} \times (d\mathbf{b})$$

$$(iii) \quad \mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$$

$$(iv) \quad (\mathbf{a} \times \mathbf{b}) \times \mathbf{c} = \mathbf{a} \times \mathbf{c} + \mathbf{b} \times \mathbf{c}$$

Theorem 8.1 (Derivative of Vector-valued Function). Let $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$ and suppose that the components f, g and h are all differentiable at $t = a$. Then $\mathbf{r}$ is differentiable at $t = a$ and its derivative is given by

$$\mathbf{r}'(a) = \langle f'(a), g'(a), h'(a) \rangle.$$

Theorem 8.2 (Derivative Rules).

Suppose $\mathbf{r}(t)$ and $\mathbf{s}(t)$ are differentiable vector-valued functions. $f(t)$ is a differentiable scalar function and c is a scalar constant. Then

- (i) $\frac{d}{dt}(\mathbf{r}(t) + \mathbf{s}(t)) = \mathbf{r}'(t) + \mathbf{s}'(t)$
- (ii) $\frac{d}{dt}(c\mathbf{r}(t)) = c\mathbf{r}'(t)$
- (iii) $\frac{d}{dt}f(t)\mathbf{r}(t) = f'(t)\mathbf{r}(t) + f(t)\mathbf{r}'(t)$
- (iv) $\frac{d}{dt}\mathbf{r}(t) \cdot \mathbf{s}(t) = \mathbf{r}'(t) \cdot \mathbf{s}(t) + \mathbf{r}(t) \cdot \mathbf{s}'(t)$
- (v) $\frac{d}{dt}(\mathbf{r}(t) \times \mathbf{s}(t)) = \mathbf{r}'(t) \times \mathbf{s}(t) + \mathbf{r}(t) \times \mathbf{s}'(t)$.

Theorem 8.3 (Arc Length Formula).

Let C be the curve given by

$$\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle, \quad a \leq t \leq b$$

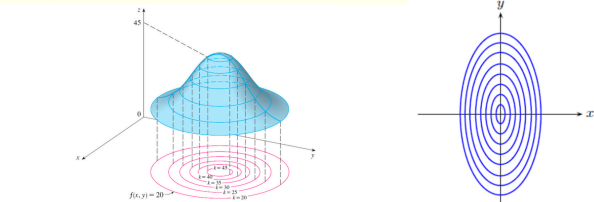
where f', g' and h' are continuous. If C is traversed exactly once as t increases from a to b , then its length is

$$s = \int_a^b \sqrt{f'(t)^2 + g'(t)^2 + h'(t)^2} dt = \int_a^b \|\mathbf{r}'(t)\| dt.$$

This is a vector tangent to the curve at $\mathbf{r}(a)$. We call $\mathbf{r}'(a)$ a **tangent vector** to the curve at $t = a$.

Definition 8.3 (Level Curve).
A **level curve** of $f(x,y)$ is the two-dimensional graph of the equation $f(x,y) = k$ for some constant k .

Definition 8.4 (Contour Plot).
A **contour plot** of $f(x,y)$ is a graph of numerous level curves $f(x,y) = k$, for representative values of k .



Example 8.7. Sketch some level curves of $h(x,y) = 4x^2 + y^2$.
Solution. If $k < 0$, then $4x^2 + y^2 = k$ has no solution, so there is no level curves for $k < 0$.
If $k = 0$, then there is only one solution $(0,0)$, so the level curve is a single point $(0,0)$.
If $k > 0$, then $4x^2 + y^2 = k$ is an ellipse:

$$\frac{x^2}{(\sqrt{k}/2)^2} + \frac{y^2}{\sqrt{k}^2} = 1.$$

Thus, larger k gives rise to an ellipse with longer major and minor axes.

Definition 8.5.
A surface is a cylinder if there is a plane P such that all the planes parallel to P intersect the surface in the same curve (when viewed in 2-dimension).

Definition 8.6 (Quadric Surface).
A **quadric surface** is the graph of a second-degree equation in three variables x, y and z :

$$Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fxz + Gx + Hy + Iz + J = 0$$

where $A, B, \dots, J$ are constants.

Definition 8.7 (Elliptic paraboloid – symmetric about the z -axis).

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z}{c}$$

Definition 8.8 (Ellipsoid). If $a = b = c$, then the ellipsoid is a sphere.

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

Definition 8.10 (Level Surface).
A **level surface** of $f(x,y,z)$ is the three-dimensional graph of the equation $f(x,y,z) = k$ for some constant k .

If the point (x,y,z) moves along a level surface, the value of $f(x,y,z)$ remains fixed.

Example 8.12. Find the level surfaces of the function

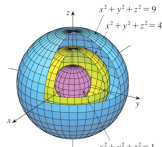
$$f(x,y,z) = x^2 + y^2 + z^2.$$

Solution. The level surfaces are:

$$x^2 + y^2 + z^2 = k$$

where $k \geq 0$.

These form a family of concentric spheres with radius $\sqrt{k}$.



Theorem 8.5 (Equation of Tangent Plane).
Suppose $f(x,y)$ has continuous first partial derivatives at (a,b) . A normal vector to the tangent plane at $(a,b,f(a,b))$ to the surface $z = f(x,y)$ is

$$\langle f_x(a,b), f_y(a,b), -1 \rangle.$$

Further, an equation of the tangent plane is given by

$$f_x(a,b)(x-a) + f_y(a,b)(y-b) - (z-f(a,b)) = 0$$

$$\text{or} \quad z = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b).$$

Theorem 8.6 (The Chain Rule - Case 1).
Suppose that $z = f(x,y)$ is a differentiable function of x and y , where $x = g(t)$ and $y = h(t)$ are both differentiable functions of t . Then, z is a differentiable function of t , and

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}.$$

Theorem 8.7 (The Chain Rule - Case 2).
Suppose that $z = f(x,y)$ is a differentiable function of x and y , where $x = g(s,t)$ and $y = h(s,t)$ are both differentiable functions of s and t . Then,

$$\frac{\partial z}{\partial s} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial s},$$

$$\frac{\partial z}{\partial t} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial t}.$$

Theorem 8.8 (The Chain Rule - General Version).
Suppose that u is a differentiable function of n variables $x_1, \dots, x_n$, and each x_j is a differentiable function of m variables $t_1, \dots, t_m$. Then u is a function of $t_1, \dots, t_m$ and

$$\frac{\partial u}{\partial t_i} = \frac{\partial u}{\partial x_1} \frac{\partial x_1}{\partial t_i} + \frac{\partial u}{\partial x_2} \frac{\partial x_2}{\partial t_i} + \dots + \frac{\partial u}{\partial x_n} \frac{\partial x_n}{\partial t_i}$$

for each $i = 1, \dots, m$.

Theorem 8.9 (Implicit Differentiation: Two Independent Variables).
Suppose the equation $F(x,y,z) = 0$, where F is differentiable, defines z implicitly as a differentiable function of x and y . Then,

$$\frac{\partial z}{\partial x} = -\frac{F_x(x,y,z)}{F_z(x,y,z)}, \quad \frac{\partial z}{\partial y} = -\frac{F_y(x,y,z)}{F_z(x,y,z)}$$

provided $F_z(x,y,z) \neq 0$.

Definition 8.13.
Let $z = f(x,y)$. Suppose Δx and Δy are increments in the independent variable x and y respectively.

Then the **increment** in z is defined by

$$\Delta z = f(x + \Delta x, y + \Delta y) - f(x, y).$$

Definition 8.14.
Let $z = f(x,y)$. Suppose Δx and Δy are increments in the independent variable x and y respectively.

Then the **differentials** of the independent variables x and y are

$$dx = \Delta x, \quad dy = \Delta y.$$

The **differential** (or **total differential**) of the dependent variable z is

$$dz = f_x(x,y)dx + f_y(x,y)dy.$$

Theorem 8.10.
Suppose f is differentiable at (a,b) . Let Δx and Δy be small increments in x and y respectively from (a,b) . Then

$$\Delta z \approx dz = f_x(a,b)dx + f_y(a,b)dy = f_x(a,b)\Delta x + f_y(a,b)\Delta y.$$

Example 8.24. The base radius and height of a circular cone are measured as 10cm and 25cm respectively, with a possible error in measurement of as much as 0.1cm in each. Use differential to estimate the maximum error in the calculated volume of the cone.

Solution. The volume of the cone is $V = \pi r^2 h/3$. So

$$dV = V_r dr + V_h dh = \frac{2\pi rh}{3} dr + \frac{\pi r^2}{3} dh.$$

Since each error is at most 0.1cm, we can take $dr = 0.1$ and $dh = 0.1$ along with $r = 10, h = 25$ to give

$$dV = \frac{500\pi}{3}(0.1) + \frac{100\pi}{3}(0.1) = 20\pi.$$

The maximum error required is $20\pi \text{ cm}^3$.

Theorem 8.11 (Computing Directional Derivative).
If $f(x,y)$ is a differentiable function, then f has a directional derivative in the direction of any unit vector $\mathbf{u} = \langle a,b \rangle$ and

$$D_{\mathbf{u}}f(x,y) = f_x(x,y)a + f_y(x,y)b.$$

We can rewrite it in terms of vectors:

$$D_{\mathbf{u}}f(x,y) = \langle f_x, f_y \rangle \cdot \langle a,b \rangle = \langle f_x, f_y \rangle \cdot \mathbf{u}.$$

Definition 8.16 (Gradient).
The **gradient** of $f(x,y)$ is the vector-valued function

$$\nabla f(x,y) = \langle f_x, f_y \rangle = f_x \mathbf{i} + f_y \mathbf{j} = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j}$$

provided both partial derivatives exist.

Theorem 8.12 (Computing 3-D Directional Derivative).

$$D_{\mathbf{u}}f(x_0, y_0, z_0) = \nabla f(x_0, y_0, z_0) \cdot \mathbf{u}$$

where

$$\nabla f = \langle f_x, f_y, f_z \rangle = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k}$$

is the gradient vector.

Theorem 8.13 (Level Surface vs ∇f).
Suppose $f(x,y)$ is differentiable function of x and y at (x_0, y_0) .

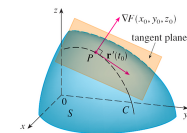
Suppose $\nabla f(x_0, y_0) \neq \mathbf{0}$. Then $\nabla f(x_0, y_0)$ is perpendicular/normal to the level curve $f(x,y) = k$ at the point (x_0, y_0) where $f(x_0, y_0) = k$.

Theorem 8.14 (Level Surface vs ∇f).
Suppose $F(x,y,z)$ is differentiable function of x, y and z at (x_0, y_0, z_0) . Suppose S is the level surface $F(x,y,z) = k$ containing (x_0, y_0, z_0) . Let C be any curve that lies on S and passes through (x_0, y_0, z_0) . Let $\mathbf{r}(t)$ be a parametric equation of C such that $\mathbf{r}(t_0) = \langle x_0, y_0, z_0 \rangle$.

Suppose $\nabla F(x_0, y_0, z_0) \neq \mathbf{0}$. Then

$$\nabla F(x_0, y_0, z_0) \cdot \mathbf{r}'(t_0) = 0,$$

That is, the $\nabla F(x_0, y_0, z_0)$ is perpendicular/normal to tangent vector $\mathbf{r}'(t_0)$ to any curve C on the surface S that passes through (x_0, y_0, z_0) .



$$\nabla F(x_0, y_0, z_0) = k, \quad \nabla F(x_0, y_0, z_0) \cdot \mathbf{r}'(t_0) = 0.$$

Theorem 8.15 (Tangent Plane to Level Surface).

$$\nabla F(x_0, y_0, z_0) \cdot \langle x - x_0, y - y_0, z - z_0 \rangle = 0$$

or equivalently,

$$F_x(x_0, y_0, z_0)(x - x_0) + F_y(x_0, y_0, z_0)(y - y_0) + F_z(x_0, y_0, z_0)(z - z_0) = 0.$$

Theorem 8.16 (Maximizing Rate of Increase/Decrease of f).
Suppose f is a differentiable function of two or three variables. Let P denote a given point. Assume $\nabla f(P) \neq \mathbf{0}$. Let $\mathbf{u}$ be a unit vector making an angle θ with ∇f . Then

$$D_{\mathbf{u}}f(P) = |\nabla f(P)| \cos \theta.$$

Moreover,

- $\nabla f(P)$ points in the direction of maximum rate of increase of f at P (maximum value of $D_{\mathbf{u}}f(P)$ is $|\nabla f(P)|$)
- $-\nabla f(P)$ points in the direction of maximum rate of decrease of f at P (minimum value of $D_{\mathbf{u}}f(P)$ is $-|\nabla f(P)|$)

Theorem 8.17.
If f has a local maximum or minimum at (a,b) and the first-order derivatives of f exist there, then

$$f_x(a,b) = f_y(a,b) = 0.$$

Definition 8.20 (Critical or Stationary Point).

Let $f(x,y) : D \rightarrow \mathbb{R}$. Then a point (a,b) is called a **critical point** of f if

- $f_x(a,b) = 0$ and $f_y(a,b) = 0$, OR
- one of the partial derivatives does not exist.

Theorem 8.18 (Second Derivative Test).
Suppose $f(x,y)$ has continuous second-order partial derivatives on some open disk centered at (a,b) . Suppose $f_x(a,b) = f_y(a,b) = 0$ (that is (a,b) is a critical point). Define the **discriminant** D for the point (a,b) by

$$D = D(a,b) = f_{xx}(a,b)f_{yy}(a,b) - [f_{xy}(a,b)]^2.$$

- (a) If $D > 0$ and $f_{xx}(a,b) > 0$, then $f(a,b)$ is a local minimum.
- (b) If $D > 0$ and $f_{xx}(a,b) < 0$, then $f(a,b)$ is a local maximum.
- (c) If $D < 0$, then (a,b) is a saddle point of f .
- (d) If $D = 0$, then no conclusion can be drawn.

critical point	D	f_{xx}	2nd Derivative Test's Conclusion
$(0,0)$	0		inconclusive
$(-1, \frac{1}{2})$	$-6 < 0$		saddle point
$(-2, 1)$	$24 > 0$	$-6 < 0$	local maximum

We need a different analysis to deal with the critical point $(0,0)$.

Notice in the plane $y = 0$, $f(x,y) = f(x,0) = x^3$. We know from Calculus that this curve has an inflection point at $x = 0$. So there is no local extremum at this point.

Moreover, when we start walking from $(0,0)$ in positive direction of x , we will be walking uphill; in the negative direction of x , we will be walking downhill.

So $(0,0)$ is a saddle point.

Theorem 9.3 (A Special Area).

$$\iint_R g(x)h(y) dA = \left(\int_a^b g(x) dx \right) \left(\int_c^d h(y) dy \right)$$

where $R = [a,b] \times [c,d]$.

$$\iint_R \frac{2x}{y} dA = \iint_R 2x \cdot \frac{1}{y} dA = \left(\int_3^4 2x dx \right) \cdot \left(\int_1^2 \frac{1}{y} dy \right)$$

$$\text{Surface area} = \iint_D dS = \iint_D \sqrt{f_x^2 + f_y^2 + 1} dA.$$

Theorem 9.9.
If $f(x,y) \geq g(x,y)$ for all $(x,y) \in D$ then

$$\iint_D f(x,y) dA \geq \iint_D g(x,y) dA.$$

Theorem 9.10. Area of plane region
Let $f(x,y) = 1$ over a given region D . Then the area of D is

$$A(D) = \iint_D 1 dA.$$

Theorem 9.11. Polar Coordinates Versus Rectangular Coordinates

$$r^2 = x^2 + y^2, \quad x = r \cos \theta, \quad y = r \sin \theta.$$

Theorem 9.12. Change to Polar Coordinates in Double Integral
If f is continuous on a polar rectangle R given by

$$R = \{(r,\theta) : 0 \leq a \leq r \leq b, \alpha \leq \theta \leq \beta\}$$

where $0 \leq \beta - \alpha \leq 2\pi$, then

$$\iint_R f(x,y) dA = \int_\alpha^\beta \int_a^b f(r \cos \theta, r \sin \theta) r dr d\theta.$$

A linear first order ODE is of the form

$$\frac{dy}{dx} + P(x)y = Q(x), \quad \text{Let } I(x) = e^{\int P(x) dx}.$$

$$y \cdot I(x) = \int Q(x) \cdot I(x) dx$$

An ODE in the form $y' + p(x)y = q(x)y^n$, where $n \neq 0, 1$, is called the Bernoulli equation. The functions $p(x)$ and $q(x)$ are continuous functions on an interval J .
Let $u = y^{1-n}$. Substituting into the Bernoulli equation we get $u' + (1-n)p(x)u = (1-n)q(x)$.

This is a first order linear ODE.

Remark. (i) When $n = 0$ or 1, the Bernoulli equation itself is a first order linear ODE.
(ii) When $n > 0$, the constant zero function $y(x) = 0$ is automatically a solution of the Bernoulli equation.

This holds for equations of the form

$$y' = g\left(\frac{y}{x}\right),$$

where g is any function of $\frac{y}{x}$. Let $v = \frac{y}{x}$. Then $y = vx$ and $y' = v' + xv'$. Then the equation $y' = g(\frac{y}{x})$ can be written as $v' + xv' = g(v)$ or equivalently $v' = \frac{g(v)-v}{x}$, which is separable. We can now solve for v , and then solve for y .
A differential equation of the form $y' = f(ax + by)$, where f is continuous and $b \neq 0$, can be solved by setting $u = ax + by$. (If $b = 0$, then the equation itself is separable.)

Summing an arithmetic sequence of n terms:

$$\sum_{k=0}^{n-1} a_k = \frac{n}{2} (2a_0 + (n-1)d)$$

Summing a geometric sequence of n terms ($r \neq 1$),

$$\sum_{k=0}^{n-1} a_k = a_0 \left(\frac{1-r^n}{1-r} \right)$$

Theorem 4.5 (FTC 2)

$$\frac{d}{dx} \int_a^{g(x)} f(t) dt = f(g(x))g'(x)$$